that the sum of the curvatures we measure is proportional to the density of mass we measure plus 3 times the pressure we measure. In this sense, the Einstein field equation is the same in every reference frame; it obeys Einstein's principle of relativity.

Under most circumstances (for example, throughout the solar system), the pressure of matter is tiny compared to its mass density times the speed of light squared, and therefore the pressure is an unimportant contributor to spacetime curvature; *the spacetime warpage is due almost solely to mass.* Only deep inside neutron stars (Chapter 5), and in a few other exotic places, is pressure a significant contributor to the warpage.

By mathematically manipulating the Einstein field equation, Einstein and other physicists have not only explained the deflection of starlight by the Sun and the motions of the planets in their orbits, including the mysterious perihelion shift of Mercury, they have also predicted the existence of black holes (Chapter 3), gravitational waves (Chapter 10), singularities of spacetime (Chapter 13), and perhaps the existence of worm-holes and time machines (Chapter 14). The remainder of this book is devoted to this legacy of Einstein's genius.

As I browse through Einstein's published scientific papers (a browsing which, unfortunately, I must do in the 1965 Russian edition of his collected works because I read no German and most of his papers have not as of 1993 been translated into English!), I am struck by the profound change of character of Einstein's work in 1912. Before 1912 his papers are fantastic for their elegance, their deep intuition, and their modest use of mathematics. Many of the arguments are the same as those which I and my friends use in the 1990s when we teach courses on relativity. Nobody has learned to improve on those arguments. By contrast, after 1912, complex mathematics abounds in Einstein's papers—though usually in combination with insights about physical laws. This combination of mathematics and physical insight, which only Einstein among all physicists working on gravity had in the period 1912–1915, ultimately led Einstein to the full form of his gravitational laws.

But Einstein wielded his mathematical tools with some clumsiness. As Hilbert was later to say, "Every boy in the streets of Göttingen understands